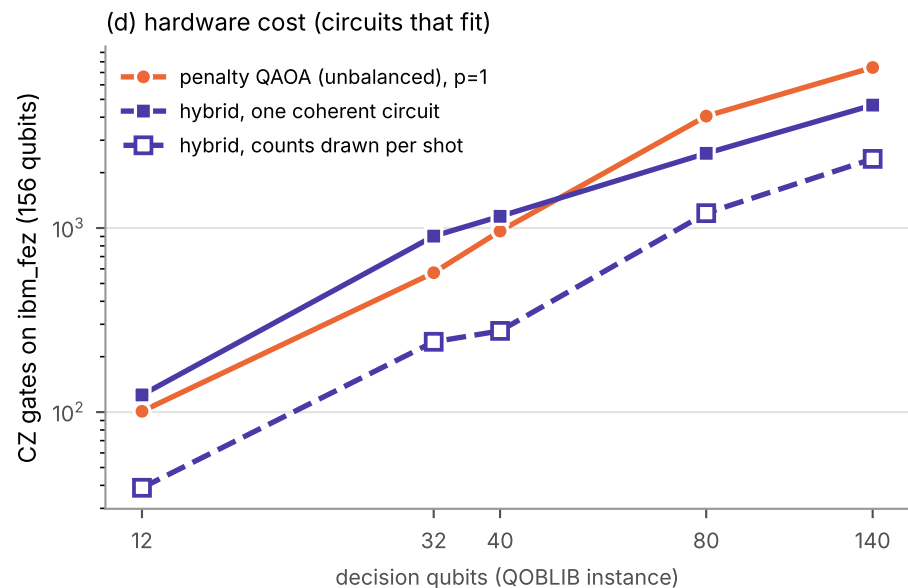
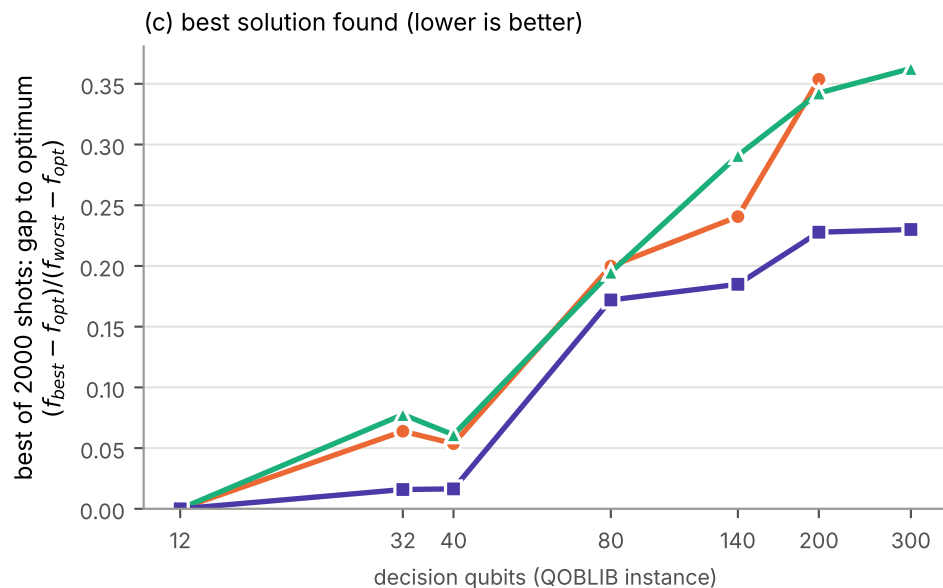
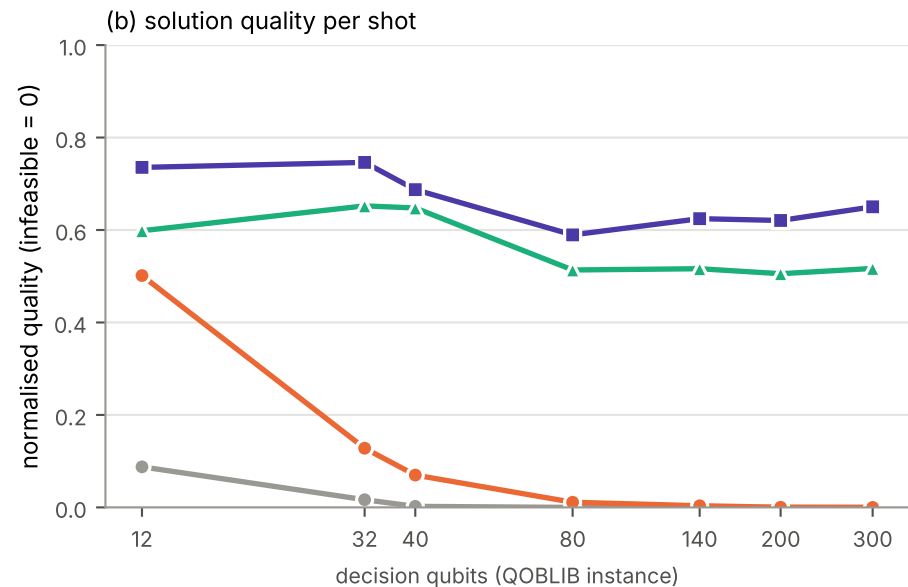
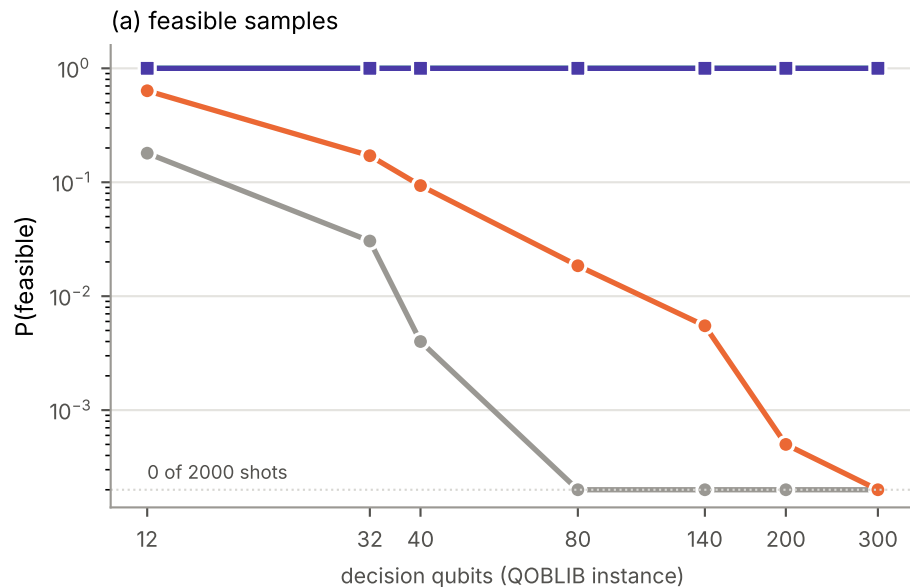




instances (decision qubits: key) — 12: a003_t02, 32: a004_t04, 40: a005_t04, 80: a010_t04, 140: a010_t07, 200: a010_t10, 300: a010_t15

Legend:

- uniform random (grey circles)
- penalty QAOA (unbalanced), $p=1$ (orange circles)
- counts from A_q + random subset (classical) (green triangles)
- hybrid: A_q counts + Dicke + XY, $p=1$ (blue squares)